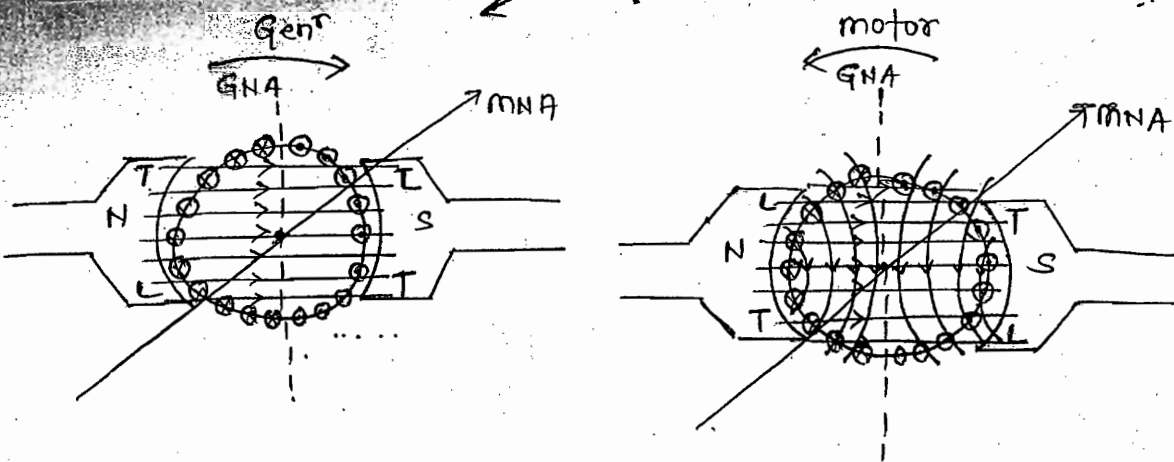


Armature Reaction in dc motors →

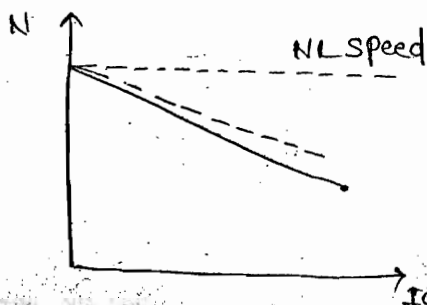


* The arm. reaction in dc motors is similar to that of genr but depending on dirn of rotation terminology will change:-

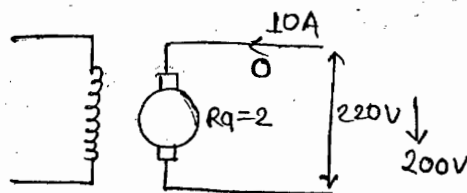
- (1) The trailing pole tips of genr becomes leading pole tips in motor (vice-versa).
- (2) The flux density will increase under the leading pole tips.
- (3) Shifting of MNA is in the opposite dirn to that of motor rotation. In order to improve commutation the brushes need to be shifted opposite to the dirn of rotation of motor. This leads to additional Dm. Consequently flux decrease, speed increase & torque decrease.
- (4) The polarity of interpole is equal to polarity of main pole behind motor rotation.

Note:- similar to genr there is a slight Dm due to magnetic saturation of poles which will increase the speed & reduce torque capability of motor.

Note:- Therefore in shunt motor with arm. reaction the speed may increase with load.



4/1



$$E_b = V - I_a R_q$$

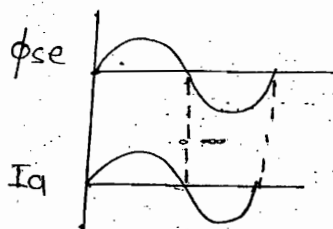
$$= 200 - 10 \times 2$$

$$= 200V$$

Because the source reduce to 200 vol. & the $E_b = 200V$ then there is no change. Hence current goes to 0 1st.

And then $N \downarrow$ $E_b \downarrow$ then it set to 10A

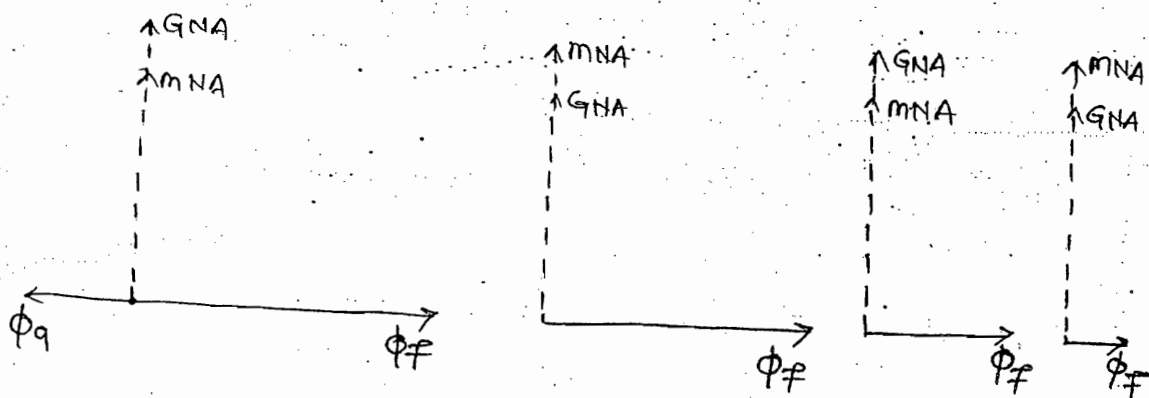
9/4



$$T \propto (\phi)(-I_a)$$

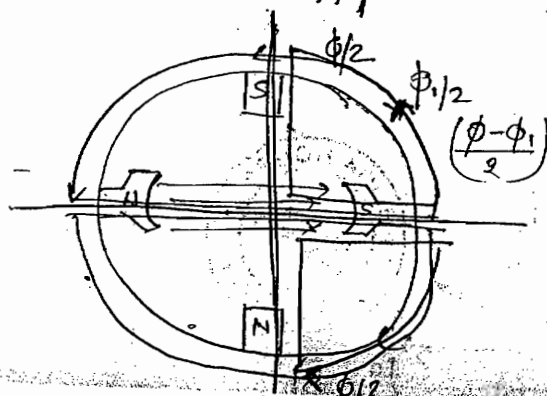
- * Because the torque is true then it can use in both.
- * But shunt motor can't operate on ac because they have huge wdg.

19/5



If there is a shifting of 90° of brush then there is only DM. & the due to DM, ϕ_f reduces. & GNA is at MNA.

20/5



51
16

$$T \propto I_a^2$$

$$T \propto I_a$$

$$\frac{T_2}{T_1} \propto \frac{I_{a2}}{I_{a1}}$$

$$\frac{T_2}{T_1} = \frac{12}{10}$$

$$T_2 = 1.2 T_1$$

20%

Series

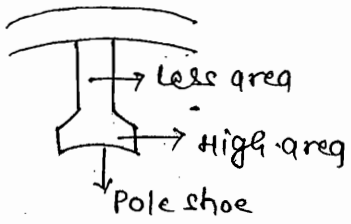
$$T \propto I_a^2$$

$$\frac{T_2}{T_1} = \left(\frac{12}{10}\right)^2$$

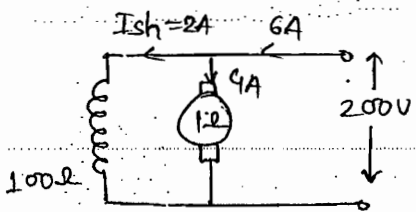
$$T_2 = 1.44 T_1$$

44%

52
16



70
13



$$C_{u loss} = I_{sh}^2 R_{sh} + I_a^2 R_a$$

$$= 4^2 \cdot 1 + 2^2 \times 100$$

$$= 416 \text{ W}$$

74
13

$$N \propto \frac{E_b}{\phi}$$

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

$$E_{b1} = V - I_a R_a = 220 - 10(1) = 210 \text{ V}$$

$$\phi_2 = 0.9 \phi_1$$

$$E_{b2} = 189$$

$$E_b = V - I_a R_a$$

$$189 = 220 - 11.11(I_a + R_{ext})$$

$$T \propto \phi I_a$$

$$\frac{T_2}{T_1} = \frac{\phi_2}{\phi_1} \cdot \frac{I_{a2}}{I_{a1}}$$

torque same

$$\phi_2 = 0.9 \phi_1$$

$$I_{a1} = 10 \text{ A}$$

$$I_{a2} = 11.11 \text{ A}$$

$$R_{ext} = 1.79 \Omega$$

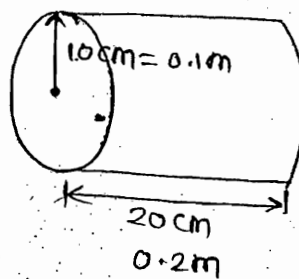
Q → A 24 slot 2p dc gen^r has 18 turns/coil with double layer wdg.
 Flux density / p 1 Tesla, length of arm, 20 cm, radius 10 cm. Poles
 are designed to cover 80% of pole pitch. If the arm, angular
 velocity is 183.2 rad/s. Determine induced emf.

Soln → 2p, 24 slots, double layer, 18T/coil, $\phi = 1 \text{ Tesla/p}$
 $l = 20 \text{ cm}$, $r = 10 \text{ cm}$, 80%, $\omega = 183.2 \text{ rad/s}$.

Calcⁿ of ϕ →
$$E_g = \frac{\phi Z N P}{60 A}$$

Total area = $2\pi r l$
 $= 0.1256 \text{ m}^2$

Total area / p = $\frac{2\pi r l}{P}$
 $= 0.0628$



Effective area under a pole = $0.8 \times \frac{2\pi r l}{P}$
 $= 0.05 \text{ m}^2$

Calcⁿ of Z → $\phi = B \cdot A = 1 \times 0.05 = 0.05 \text{ wb}$

No. of coil = 24 (double layer).

Total no. of turns = $18 \times 24 = 432 \text{ turns}$.

No. of Cond^r = $432 \times 2 = 864$

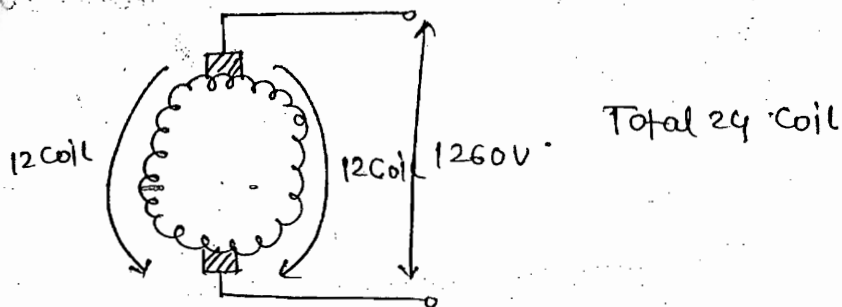
Calcⁿ of N →

$N = \frac{60}{2\pi} \times 183.2 = 1750 \text{ rpm}$

$$E_g = \frac{\phi Z N P}{60 A} = \frac{0.05 \times 864 \times 1750 \times 2}{60 \times 2}$$

$$E_g = 1260 \text{ V}$$

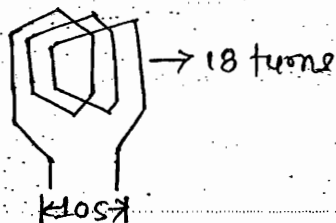
~ Induced emf per parallel path



12 coils in each path are producing 1260V as the coils are in series.

$$\text{voltage per coil} = \frac{1260}{12} = 105\text{V}$$

Because of 18 turn coil



$$\text{Voltage / turn} = 105 / 18 = 5.83\text{V}$$



$$\text{Voltage / cond} = 5.83 / 2 = 2.9\text{V}$$

Q. → A 4P lap wound dc sh. gen^r arm. wdg. consist of 220 turns. Each of resistance 0.004. what is the arm. resistance.
solⁿ →

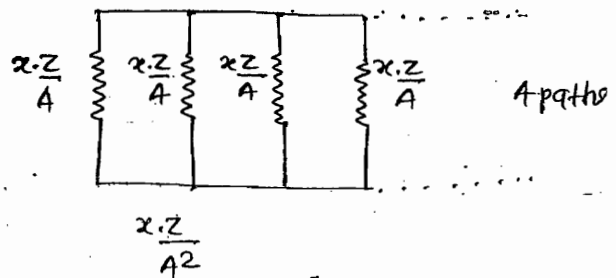
$$R_a = x \cdot \frac{Z}{A^2}$$

$$220 \text{ turns} = 440 \text{ cond}^r$$

$$0.004 \Omega \text{ / turn} = 0.002 \Omega \text{ cond}^r$$

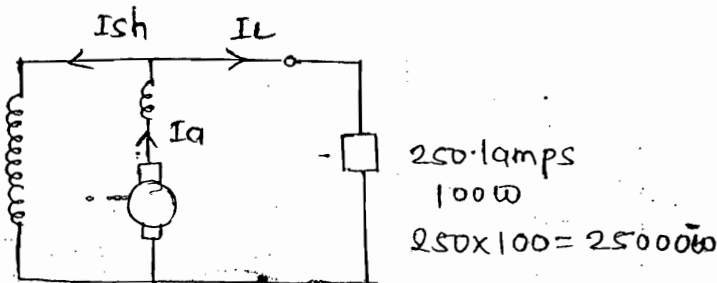
$$R_a = 0.002 \times \frac{440}{4^2}$$

$$R_a = 0.055 \Omega$$



Ques → A Compound gen^r is to supply 250 lamps each rated 100w at 250 Volt. Arm. resistance 0.06, series resistance 0.042, $R_{sh} = 50\Omega$ determine the generated emf when m/c is connected w^l long & short shunt (BCD = 1V/brush).

Soln →



$$I_L = \frac{25000}{250} = 100A \quad I_{sh} = \frac{250}{50} = 5A$$

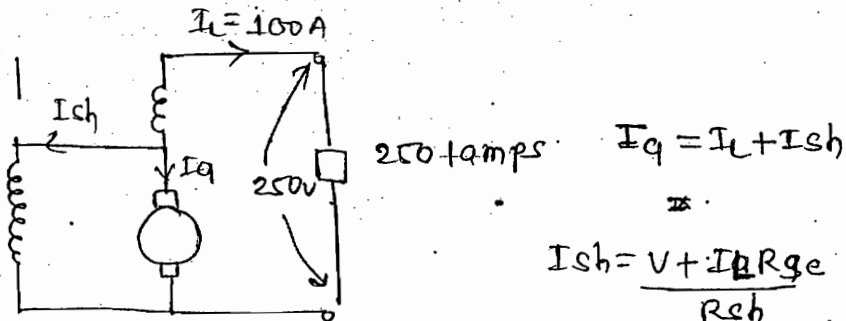
$$I_a = I_L + I_{sh} = 100 + 5$$

$$I_a = 105A$$

$$E_g = V + I_a(R_a + R_{se}) + BCD$$

$$= 250 + 105(0.06 + 0.04) + 2$$

$$E_g = 262.5V$$



$$I_a = I_L + I_{sh}$$

$$I_{sh} = \frac{V + I_a R_{se}}{R_{sh}}$$

$$= \frac{250 + 100(0.04)}{50}$$

$$= 5.08A$$

$$I_a = I_L + I_{sh}$$

$$= 100 + 5.08$$

$$I_a = 105.08A$$

$$E_g = V + I_a R_a + I_a R_{se} + BCD$$

$$= 250 + 105.08(0.06) + 100(0.04) + 2 \times 10$$

$$E_g = 262.3V$$

Q. → In a long shunt compound gen^r the terminal vol. is 230 when the gen^r delivers 150A. Determine the power generated & distribution of this power.

Solⁿ: Given → $V = 230V$ $R_{se} = 0.015\Omega$

$$I_L = 150A$$

$$R_a = 0.032\Omega$$

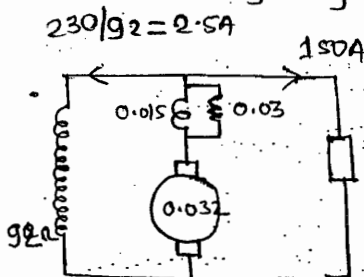
$$R_{sh} = 92\Omega$$

$$R_d = 0.03\Omega$$

& a diverter of resistance 0.03 is connected.

Solⁿ →

$$P_g = E_g I_a, \quad P_d = V \cdot I_L$$



$$R_d \parallel R_{se} = \frac{0.015 \times 0.03}{0.015 + 0.03}$$

$$= 0.01\Omega$$

$$E_g = V + I_a R_a + I_a (0.01)$$

$$= 230 + 152.5(0.032) + 152.5(0.01)$$

$$= 236.405V$$

Power generated in arm $P_g = E_g I_a$

$$= 236.405 \times 152.5$$

$$= 36051.762W$$

Total cu loss →

$$① I_a^2 R_a = (152.5)^2 (0.032) = 144.2W$$

$$* V \cdot I_L = 230 \times 150$$

$$= 34500W$$

$$② I_{sh}^2 R_{sh} = (2.5)^2 (92) = 575W$$

$$③ I_a^2 (R_d \parallel R_{se}) = (152.5)^2 (0.01) = 232.56W$$

$$P_g - TCL = P_{delivered} = 152.5 \times 36051.762 - 1551.76$$

$$= 34500.002W$$

Que. → A 220V dc sh. motor takes a current of 80A runs at 800 rpm

$$I_L = 80A, N = 800 \text{ rpm}, R_{sh} = 50\Omega, R_a = 0.1\Omega$$

$$P_{rot \text{ loss}} = 1600W$$

- ① Cu loss ② T_q ③ T_{sh} ④ η

Soln →

①

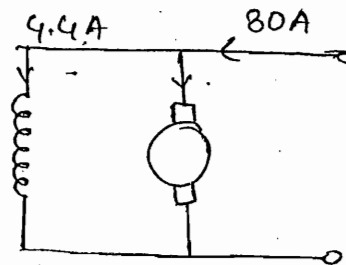
$$I_a = I_{sh} \quad I_{sh} = \frac{V}{R_{sh}}$$

$$I_a + I_{sh} = I_L$$

$$I_a = I_L - I_{sh}$$

$$= 80 - 4.4$$

$$I_a = 75.6A$$



$$* Cu \text{ loss} = I_a^2 R_a + I_{sh}^2 R_{sh}$$

$$= 1539.536W$$

②

$$T_q = \frac{60}{2\pi N} E_b I_a$$

$$= \frac{60}{2\pi \times 800} E_b I_a$$

$$E_b = V - I_a R_a = 220 - 75.6 \times 0.1 = 212.44V$$

$$T_q = \frac{60}{2\pi \times 800} \times 212.44 \times 75.6$$

$$T_q = 191.767 \text{ N-m}$$

$$* V I_L - E_b I_a = 220 \times 80 - 16060.464$$

$$= 1539.536W$$

$$T_{sh} = \frac{60}{2\pi N} (E_b I_q - \text{Rot. loss})$$

$$= \frac{60}{2\pi \times 800} (212.4 \times 75.6 - 1600)$$

$$= 172.6 \text{ N-m}$$

④

$$\eta = \frac{o/p}{i/p} = \frac{P_{sh}}{V I_L} = \frac{(212.4 \times 75.6 - 1600)}{220 \times 80}$$

$$\eta = 82.14\%$$

Q. → An 8P dc gen^r has 480 arm. cond^r with wave wdg. Arm. current 200A. Find arm. reaction Dm & cm amp-turns/p. If ① The brushes are on GNA.

② Brushes are shifted by 6° E from GNA.

③ Brushes are shifted by 6° mech. by GNA.

solⁿ →

$$\text{Arm. amp.-turns/p.} = \frac{I_a Z}{A \cdot 2P} = \frac{200}{2} \times \frac{480}{2 \times 8}$$

$$= 3000 \text{ AT/p}$$

① For $\theta = 0$ (Brush are on GNA)

$$A_d/p = \frac{20}{180} \times \frac{I_a}{A} \times \frac{Z}{2P} = 0$$

$$A_{tc}/p = \frac{180-20}{180} \times 3000 = 3000 \text{ AT/p}$$

②

$$\theta = 6^\circ$$

$$A_d/p = \frac{6 \times 2}{180} \times 3000 = 200 \text{ AT/p}$$

$$A_{tc}/p = \frac{180-6 \times 2}{180} \times 3000 = 2800 \text{ AT/p}$$

③

$$\theta_e = \frac{P}{2} \theta_m$$

$$= \frac{3}{2} \times 6 = 24^\circ \text{E}$$

$$\therefore AT_d/p = \frac{2 \times 24}{180} \times 3000 = 800 \text{ AT/p}$$

$$AT_c/p = \frac{180 - 2 \times (24)}{180} \times 3000 = 2200 \text{ AT/p}$$

Q. → In 8p lap wound dc m/c has 720 arm, Cond'r. Pole covers 70% of pole pitch. Cal. the no. of Cond'r of cw under each pole.

Soln →

$$Z_c = \frac{Z}{A_p} \times \left(\frac{\text{Pole arc}}{\text{Pole pitch}} \right)$$

$$= \frac{720}{8 \times 8} (0.7)$$

$$Z_c = 8$$

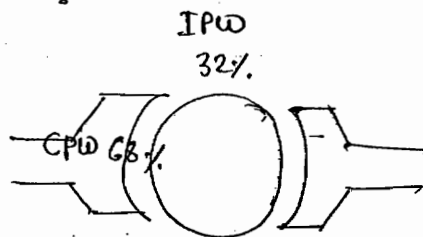
Q. → A dc m/c has arm. amp. turns/p (AT/p) 15000, ratio of pole arc / pole pitch 0.68, interpolar airgap length 1cm. & additional flux density required 0.25 T/p. Rated arm current 850. Cal. cw. turn/p & IPW turn/p.

Soln → cw AT/p = 0.68 (15000)

$$= 10200$$

$$IPW AT/p = 0.32 (15000)$$

$$= 4800$$



$$\text{flux} = \frac{\text{MMF}}{\text{Reluctance}}$$

To calculate turns $\frac{AT}{A} = T$

$$\text{CW } AT/p = AT/p / I_a = \frac{0.68(15000)}{850} = 12 \text{ turns}$$

$$\text{IPW } AT/p = \frac{4800}{850} = 6 \text{ turns}$$

$$\phi = \frac{AT}{\frac{l}{\mu_0 \mu_r}}$$

$$\mu_r = 1, \mu_0 = 4\pi \times 10^{-7}$$

$$AT = \frac{\phi l}{\mu_0 \mu_r}$$

$$\boxed{\frac{Bl}{\mu_0} = AT}$$

$$\text{Additional turns req. } AT = \frac{0.25 \times 0.01}{4\pi \times 10^{-7}} = 1989 \text{ AT/p}$$

$$\text{IPW } AT/p = 4800 + 1989$$

$$= 6789$$

$$\text{IPW } AT/p / T = \frac{6789}{850} = 8 \text{ turns}$$

(48/9)

As brush shift angle $\alpha \uparrow$ interaction angle b/w the field decrease. Consequently torque decrease

$$- \alpha \uparrow T \downarrow$$

$$q\eta s \rightarrow \cos \alpha$$